Exercise 1.2: Amdahl's Law

1. Derivation

Let T_1 be the sequential execution time of a fixed workload, P the fraction of this time that can be parallelized, and N the number of processors. The serial part takes $(1-P)T_1$. Assuming perfect load balancing, no parallel overhead, and an ideal N-fold acceleration of the parallel part, its execution time is PT_1/N. Hence

$$T_N = (1-P)T_1 + \frac{PT_1}{N} = T_1\left((1-P) + \frac{P}{N}\right). \tag{1}$$

The overall speedup is therefore

$$\boxed{S(N) = \frac{T_1}{T_N} = \frac{1}{(1-P) + P/N}}. \tag{2}$$

For $0 \leq P < 1$,

$$\lim_{N\to\infty} S(N) = \frac{1}{1-P}. \tag{3}$$

The serial fraction sets the asymptotic limit. For the special case $P = 1$, the ideal speedup is $S(N) = N$ and there is no finite serial-fraction limit.

2. Correctness for real applications

Amdahl's law is correct under its assumptions. It is not a universal prediction for every parallel execution: using N as the acceleration of the parallel portion assumes that overhead and other architectural effects can be neglected.

Performance predicted too high. Communication, synchronization, thread creation, and load imbalance increase the actual time. Consider a computation with an ideal 100× speedup whose processors exchange intermediate results over a network. If communication becomes a bottleneck, the actual speedup is below 100×. Writing

$$T_N = T_1\left((1-P) + \frac{P}{N}\right) + T_{\text{overhead}}(N)$$

makes clear why positive overhead reduces the ideal speedup.

Performance predicted too low. A working set that exceeds the cache of one processor may fit into the aggregate caches when partitioned across several processors. Reducing slow memory accesses can accelerate the parallel part by more than N, so actual speedup can exceed the prediction of Equation (2), and can even be superlinear. This changes the assumed acceleration of the parallel portion; it does not invalidate the time-decomposition argument.

Changing the workload. In weather forecasting, more processors are often used to increase grid resolution rather than to finish an unchanged problem sooner. Amdahl's fixed-workload limit does not directly characterize this comparison. Weak scaling studies increasing the workload in proportion to the number of processors, while Gustafson's law addresses scaled speedup using the serial fraction of a parallel run. These perspectives change the workload assumptions used in the fixed-size Amdahl model.

3. Numerical calculations and comparison

(a) $P = 0.5$ and $N = 4$

$$S_A = \frac{1}{(1-0.5)+0.5/4} = \frac{1}{0.5+0.125} = \boxed{1.6}.$$

The parallel execution takes $0.625T_1$, and its efficiency is $S_A/4 = 0.4 = 40\%$.

(b) $P = 0.2$ and $N = 8$

$$S_B = \frac{1}{(1-0.2)+0.2/8} = \frac{1}{0.8+0.025} = \boxed{\frac{40}{33} \approx 1.212}.$$

The parallel execution takes $0.825T_1$, and its efficiency is $S_B/8 = 5/33 \approx 15.15\%$.

(c) Comparison

Case	P	N	Speedup	Limit as $N \to \infty$
A	0.5	4	1.6	2
B	0.2	8	1.212...	1.25

Case A is faster relative to its sequential baseline despite having fewer processors. Its larger parallelizable fraction leaves a smaller serial bottleneck. In case B, 80% of the original time remains serial, so even arbitrarily many processors cannot produce a speedup above 1.25.

These cases change both P and N, so their difference cannot be attributed to processor count alone. For any fixed $P > 0$, adding processors increases ideal speedup, but with diminishing returns when $P < 1$. Reducing the serial fraction is essential for effective scaling.

For comparison, keeping $P = 0.5$ and increasing only N from four to eight would give

$$S(8) = \frac{1}{0.5+0.5/8} = \frac{16}{9} \approx 1.778.$$